Serat Mahmud Saad

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Research Interests

Observational Astronomy: Stellar Astrophysics, AGN and Quasars, Dynamics
Computational Techniques: Machine Learning, AI, Quantum Computing

Education

Ph.D. in Astrophysics (in progress), The Ohio State University — 2025-Present
Advisor: Prof. Yuan-Sen Ting

A.B. in Physics & Applied Mathematics, Vanderbilt University — 2021-2025
Thesis: Understanding Accretion Activity in Young Stars using SDSS-V BOSS Spectra
Advisor: Prof. Keivan Stassun

Honors & Awards

University Distinguished Fellowship, The Ohio State University	2025-2026
University Fellowship, The Ohio State University	2025
Larry R. Cathey Award, Vanderbilt University	2025
Hult Prize, National Finalist	2026
Highest Honors in Astronomy, Vanderbilt University	2025
International Full-ride Student Grant, Vanderbilt University	2021-2025
Dean's List, Vanderbilt University	2021-2025
Pi Mu Epsilon, Vanderbilt University	2024
Sigma Pi Sigma, Vanderbilt University	2023
Buchanan Fellowship, Vanderbilt University	2022
Silver Medal, International Olympiad on Astronomy and Astrophysics	2020

First Author Publications

1. **Serat M. Saad**, D. M. Rowan, K. Z. Stanek et al. *PANTERA. I. Hunting for the Most Metal-Rich Stars in the Solar Neighborhood with High-Resolution Spectroscopy.* July, 2026. Submitted to OJAp. Doi: arXiv.2607.27328

2. **Serat M. Saad** & Yuan-Sen Ting. *Gravitational Anomaly Measurement in Wide Binaries is Sensitive to Orbital Modeling.* March, 2026. Submitted to OJAp. Doi: arXiv.2603.11015

3. **Serat M. Saad** & Yuan-Sen Ting. *High-Precision Differential Radial Velocities of C3PO Wide Binaries: A Test of Modified Newtonian Dynamics (MOND).* January, 2026. Submitted to OJAp. Doi: arXiv.2512.19652

4. **Serat M. Saad**, Marina Kounkel, Keivan G. Stassun et al. *ABYSS III: Observing accretion activity in young stars through empirical veiling measurements.* June, 2025. Published by AJ. Doi: 10.3847/1538-3881/ade43b

5. **Serat M. Saad**, Kaitlyn Lane, Marina Kounkel, Keivan G. Stassun et al. *ABYSS II: Identification of young stars in optical SDSS spectra and their properties.* January, 2024. Published by AJ. Doi: 10.3847/1538-3881/ad2001

Co-Author Publications

1. SDSS Collaboration et al. including **Serat M. Saad**. *The Twentieth Data Release of the Sloan Digital Sky Surveys.* July, 2026. Doi: arXiv:2607.26149
2. Zimo Cheng, Sharon X. Wang, Fan Liu et al. including **Serat M. Saad**. *A Bayesian Search for Planet Engulfment Signatures in Solar Analogs.* July, 2026. Submitted to AAS Journals. Doi: arXiv:2607.03504
3. SDSS Collaboration et al. including **Serat M. Saad**. *The Nineteenth Data Release of the Sloan Digital Sky Surveys.* June, 2026. Published by ApJS. Doi: 10.3847/1538-4365/ae4697
4. Yuan-Sen Ting, **Serat M. Saad**, Fan Liu et al. *Egent: An Autonomous Agent for Equivalent Width Measurement.* May, 2026. Published by OJAp. Doi: 10.33232/001c.161879

Telescope Timing Proposals

1. *Principal Investigator*, Large Binocular Telescope (PEPSI), 2026B. "PANTERA: Precise Abundances of the Nearest, Most Extreme Metallicity Stars." (20 hours)
2. *Co-Investigator*, Automated Planet Finder (Levy/APF), Lick Observatory, 2026B. "Characterizing the most metal-rich stars in the Solar neighborhood." Principal Investigator: D. M. Rowan. (40 hours)
3. *Principal Investigator*, Large Binocular Telescope (PEPSI), 2026A. "PASTA Survey: High precision stellar chemical abundance for 36 planet-hosting stars."
4. *Principal Investigator*, Large Binocular Telescope (PEPSI), 2025B. "PASTA Survey: High precision stellar chemical abundance for 43 planet-hosting stars."
5. *Co-Investigator*, Chandra X-ray Observatory, Cycle 26, 2024. General Observing. "Pink Mergers: Dwarf Galaxies with Extremely Luminous Quasars." Principal Investigator: Prof. Adi Foord.

Conference Posters and Abstracts

1. **Serat M. Saad**, Jessie C. Runnoe, Michael Eracleous. *An Alternative to the Supermassive Black Hole Binary Hypothesis for J1430 (Tick Tock).* AAS Winter Conference 2025. Doi: 2025aas/24535502S
2. **Serat M. Saad**, Marina Kounkel, Keivan G. Stassun. *Using low resolution optical spectra to identify young stars and their properties.* Cambridge Cool Stars Conference 2024. Doi: 10.5281/zenodo.13007729
3. **Serat M. Saad**, Anish Giri, Mubarak Ganiyu, David Nizovsky et al. *Developing Workforce in Quantum Industry: The Wond'ry Qunatum Studio.* IEEE International Conference on Quantum Computing and Engineering (QCE) 2023. Doi: 10.1109/QCE57702.2023.10282

Teaching Experiences

TA, DS 2100: Statistics for data science., Vanderbilt University	Spring 2025
TA, ASTR 1010: Introductory Astronomy, Vanderbilt University	Spring 2024
Mentor, Bangladesh Olympiad on Astronomy and Astrophysics	2021-2024
Grader, DS 1100: Introduction to data science, Vanderbilt University	Fall 2023
Instructor, Quantum Computing for everyone, TWQS Vanderbilt	Fall 2023
Grader, CS 1100: Introduction to problem solving with python.	2022-2023

Tutor, Vanderbilt Student Athletics 2022

Other Experiences

Co-founder, porAI	2026-
Author, Astrobites	2026-
Academic Team Member, Bangladesh Olympiad on Astronomy and Astrophysics	2021 - 2024
Co-lead, The Wond'ry Qunatum Studio	2022 - 2023
Team Leader, International Olympiad on Astronomy and Astrophysics	2022

Non-Refereed Publications

1. **Serat M. Saad**. *Cosmic Cannibalism: When Stars Eat Their Planets*. Astrobites, May 2026.
2. **Serat M. Saad**. *Wormholes Might Be More Real Than We Thought*. Astrobites, April 2026.
3. **Serat M. Saad**. *A Wandering Supermassive Black Hole Eating a Star*. Astrobites, March 2026.
4. **Serat M. Saad**. *Pixels and Persuasion*. Section in the digital exhibition *As Seen on TV: American Culture Through Advertisements*, Jean and Alexander Heard Libraries, Vanderbilt University.